IA: STATISTICAL LEARNING AND PREDICTION

Group Work

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GROUP WORK: Document Classification and Information Extraction

The goal of this project is to develop a **technical solution capable of classifying documents into different categories and extracting specific information from invoices**, using traditional Artificial Intelligence techniques and data processing methods, without employing generative AI models.

- Project Requirements:
  1. Document Classification:
     - Design and implement a system that classifies documents into **at least four (4) different categories**. **One of these categories must be invoices.**
     - The other three categories can be freely chosen by each group (e.g., contracts, technical reports, emails, receipts, etc.).
     - Justify the selection of the chosen categories and describe the key characteristics that enable their differentiation.

  2. Information Extraction:
     - The system must automatically extract structured information only from documents classified as invoices.
     - The minimum data to be extracted should include:
       - Invoice number
       - Invoice date
       - Due date
       - Issuer name
       - Recipient name
       - Total amount

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- Rules:
  - The use of **generative AI is not allowed**
  - Traditional text classification techniques, pattern analysis, rule-based approaches with regular expressions, classical Machine Learning models (SVM, Random Forest, etc.), and traditional Natural Language Processing (NLP) and/or Computer Vision (CV) methods should be used.

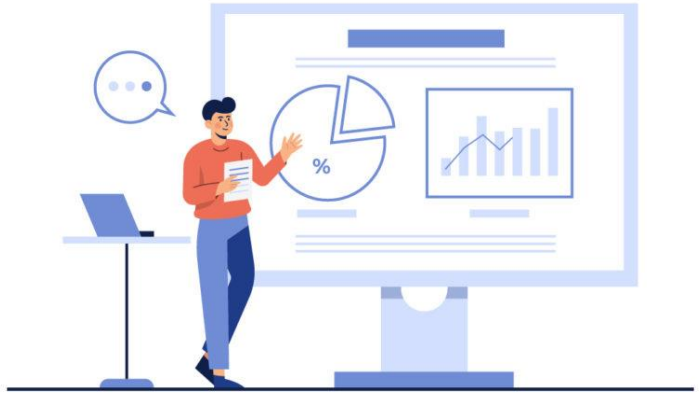
- Project Deliverables:
  - **Technical documentation**: Describe the approach used, design decisions, classification and extraction methods implemented, as well as the results obtained.
  - **End-to-end Solution documentation**
  - **Source code**: Including clear instructions for running the system.
  - **Oral presentation (15min)**: Each group will present their solution, explaining the approach, challenges encountered, and implemented solutions.
  - **Live demonstration**: At the end of the oral presentation, each group must conduct a live demonstration of their solution, showcasing its functionality with practical examples.

- Evaluation Criteria:
  - Accurate classification of different document categories.
  - Precision and completeness in the extraction of information from invoices.
  - Technical justification of the approach used.
  - Quality of documentation and clarity in the oral presentation.
  - Quality and effectiveness of the live demonstration.
  - Creativity and innovation in the proposed solution.

GROUPS

- Group Formation:
  - 8 groups will be created randomly.

- Grading Criteria:
  - The final grade **for each student** will be a weighted average of two factors:
    - **Group's Final Grade**: Based on the project deliverables, presentation, and demonstration.
    - **Peer Evaluation**: Each student must complete, **before group work deadline**, an anonymous survey rating the performance of each of their teammates on a scale from 1 to 10. **All grades should be different and don't grade yourself.**
    - If a student decides not to evaluate his or her classmates, he/she will receive a 0/10 as an individual grade.
    - Evaluation Link
    - I you decide not to rate, be aware you'll get 0/10 in your peer evaluation.



- Presentations order:
  - Will be randomly defined.

THANK YOU